QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

aptos-core @ 81d89f428fe8 · Aptos Labs — public repository (passive review)

Organization	QSI
Report date	2026-11-06 16:20 UTC
Report version	1
Audit run	30269b00-7b13-4dea-be58-d53a0e2a5767 · Document audit · 2026-11-06 16:20 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`496bbf2c0ceaa6407d995b8152ee8fd44f8f95ed1f641fbd08068a5bb313751f`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

9	0	0	9	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	9
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-19659	Medium	Quantum-vulnerable BLS12-381: transaction and message signing (2 file(s)) preliminary	35.6
QSI-2026-19660	Medium	Quantum-vulnerable Ed25519: transaction and message signing (10 file(s)) preliminary	35.6
QSI-2026-19662	Medium	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (2 file(s)) preliminary	35.6
QSI-2026-19663	Medium	Quantum-vulnerable ECDSA: transaction and message signing (3 file(s)) preliminary	35.6
QSI-2026-19667	Medium	Quantum-vulnerable BN254 (alt_bn128): transaction and message signing (1 file(s)) preliminary	35.6

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 2 of 4 — post-quantum in use outside the critical paths. Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

Scope and boundaries

Mode: Document audit

Repository: https://github.com/aptos-labs/aptos-core @ 81d89f428fe8e4b69d53b4916993e9a930afc37c — evidence links open the lines at this commit.

Supplied inputs

336 files, 3667051 bytes in total, 8 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	331	aptos-move/framework/src/aptos.rs: 223 lines scanned with 16 rule patterns and 267 registry patterns; aptos-move/framework/src/built_package.rs: 773 lines scanned with 16 rule patterns and 267 registry patterns; aptos-move/framework/src/chunked_publish.rs: 223 lines scanned with 16 rule patterns and 267 registry patterns
dependencies	checked	4	crates/aptos-crypto/Cargo.toml: 78 dependencies parsed (0 pinned); keyless/pepper/common/Cargo.toml: 18 dependencies parsed (0 pinned); keyless/pepper/service/Cargo.toml: 36 dependencies parsed (0 pinned)
secrets	checked	3	crates/aptos-crypto/src/x25519.rs: 2 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name); types/src/network_address/mod.rs: value(s) at line(s) 908 (test_network_address_display), 1069 (test_find_noise_proto), 1089 (test_parse_handshake) are declared inside a fixture and treated as test data, although the file itself is not test code; types/src/network_address/mod.rs: 5 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)
keys	checked	1	types/src/jwks/rsa/insecure_test_jwk_private_key.txt: private key block at line 1 is not reported: its path marks it as a non-production key ("insecure")
certificates_and_keys	checked	1	types/src/jwks/rsa/insecure_test_jwk_private_key.txt: certificates/keys parsed
entropy	checked	1	random-number generation: 331 code files read with 0 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	not checked	1	no exactly pinned versions to query

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 337 (input files: C)

Not checked

- osv_advisories: no exactly pinned versions to query

Architecture overview

Static review of the source tree aptos-core at 81d89f428fe8e4b69d53b4916993e9a930afc37c. 336 files analysed; of 625 files, 95 are test, fuzzing, benchmark or example code and were left out, 194 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. The review was restricted to 5 path(s) of the repository: aptos-move/framework/aptos-framework/sources, aptos-move/framework/src, crates/aptos-crypto, keyless, types; 22057 files outside them were not read. Files no analyzer reads include: aptos-move/framework/aptos-framework/sources/account/account.move, aptos-move/framework/aptos-framework/sources/account/account.spec.move, aptos-move/framework/aptos-framework/sources/account/account_abstraction.move, aptos-move/framework/aptos-framework/sources/account/account_abstraction.spec.move, aptos-move/framework/aptos-framework/sources/account/auth_data.move. No requests were sent to the owner's systems.

Overall security posture

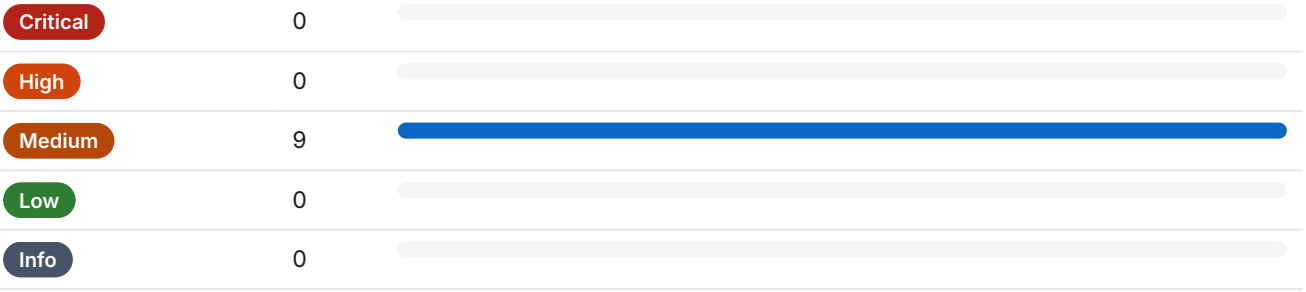
100
Indicative posture index (0–100)

60.4%
Quantum-vulnerable share of inventory

96
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-19659	Quantum-vulnerable BLS12-381: transaction and message signing (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19660	Quantum-vulnerable Ed25519: transaction and message signing (10 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19662	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19663	Quantum-vulnerable ECDSA: transaction and message signing (3 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19667	Quantum-vulnerable BN254 (alt_bn128): transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19661	Quantum-vulnerable X25519: role not determined (3 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19664	Quantum-vulnerable Ed25519: role not determined (4 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19665	Quantum-vulnerable BN254 (alt_bn128): role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19666	Quantum-vulnerable RSA: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary

Detailed findings

QSI-2026-19659 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BLS12-381: transaction and message signing (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to BLS12-381 across 2 file(s), e.g. crates/aptos-crypto/src/bls12381/bls12381_keys.rs:77. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:BLS12-381:transaction_signing
Data source	crates/aptos-crypto/src/bls12381/bls12381_keys.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/aptos-crypto/src/bls12381/bls12381_keys.rs:77 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/aptos-crypto/src/bls12381/bls12381_keys.rs · line 77 · SHA-256 f86b41febe322eda… · File b2ebf8767ada59dc…

```
   76 |     pub fn aggregate(pubkeys: Vec<&Self>) -> Result<PublicKey> {
   77 |         let blst_pubkeys: Vec<_> = pubkeys.iter().map(|pk| &pk.pubkey).collect();
   78 |
```

file_excerpt · crates/aptos-crypto/src/bls12381/bls12381_keys.rs · line 80 · SHA-256 a99c05717793c659… · File b2ebf8767ada59dc…

```
   79 |         // CRYPTONOTE(Alin): We assume the PKs have had their PoPs verified and thus have
also been subgroup-checked
   80 |         let aggpk = blst::min_pk::AggregatePublicKey::aggregate(&blst_pubkeys[..], false)
   81 |             .map_err(|e| anyhow!("{:?}", e))?;
```

file_excerpt · crates/aptos-crypto/src/constant_time/zkcrypto_scalar_mul.rs · line 4 · SHA-256 221f1dca25f14b9d… · File c791efc464274512…

```
    3 |
    4 | use bls12_381::{G1Projective, Scalar};
    5 | use dudect_bencher::{
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19660 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: transaction and message signing (10 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

151 reference(s) to Ed25519 across 10 file(s), e.g. crates/aptos-crypto/src/ed25519/ed25519_keys.rs:9. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	crates/aptos-crypto/src/ed25519/ed25519_keys.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/aptos-crypto/src/ed25519/ed25519_keys.rs:9 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_keys.rs · line 9 · SHA-256 adf76b09b67a86ab… · File 79a6f426fb5efb58…

```
  8 | use crate::{
  9 |     ed25519::{Ed25519Signature, ED25519_PRIVATE_KEY_LENGTH, ED25519_PUBLIC_KEY_LENGTH},
 10 |     hash::CryptoHash,
```

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_keys.rs · line 16 · SHA-256 1caccba153f57a75… · File 79a6f426fb5efb58…

```
15 | use curve25519_dalek::{edwards::CompressedEdwardsY, scalar::Scalar};
16 | use ed25519_dalek::ExpandedSecretKey;
17 | #[cfg(any(test, feature = "fuzzing"))]
```

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_keys.rs · line 24 · SHA-256 1a0100275641151d... · File 79a6f426fb5efb58...

```
23 | #[derive(DeserializeKey, SerializeKey, SilentDebug, SilentDisplay)]
24 | pub struct Ed25519PrivateKey(pub(crate) ed25519_dalek::SecretKey);
25 |
```

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_sigs.rs · line 7 · SHA-256 e348dba0532f4e56... · File 838e7e95fe58782e...

```
 6 | use crate::{
 7 |     ed25519::{Ed25519PrivateKey, Ed25519PublicKey, ED25519_SIGNATURE_LENGTH, L},
 8 |     hash::CryptoHash,
```

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_sigs.rs · line 19 · SHA-256 7c9df4c2d607763e... · File 838e7e95fe58782e...

```
18 | #[derive(DeserializeKey, Clone, SerializeKey)]
19 | pub struct Ed25519Signature(pub(crate) ed25519_dalek::Signature);
20 |
```

file_excerpt · crates/aptos-crypto/src/ed25519/ed25519_sigs.rs · line 44 · SHA-256 2d4aff20c28e0074... · File 838e7e95fe58782e...

```
43 |     ) -> std::result::Result<Ed25519Signature, CryptoMaterialError> {
44 |         match ed25519_dalek::Signature::try_from(bytes) {
45 |             Ok(dalek_signature) => Ok(Ed25519Signature(dalek_signature)),
```

file_excerpt · crates/aptos-crypto/src/ed25519/mod.rs · line 34 · SHA-256 4109af929d07df4d... · File 7b21ea56b9c4fbf3...

```
33 | /// The length of the Ed25519PrivateKey
34 | pub const ED25519_PRIVATE_KEY_LENGTH: usize = ed25519_dalek::SECRET_KEY_LENGTH;
35 | /// The length of the Ed25519PublicKey
```

file_excerpt · crates/aptos-crypto/src/ed25519/mod.rs · line 36 · SHA-256 e36e845052f25c99... · File 7b21ea56b9c4fbf3...

```
35 | /// The length of the Ed25519PublicKey
36 | pub const ED25519_PUBLIC_KEY_LENGTH: usize = ed25519_dalek::PUBLIC_KEY_LENGTH;
37 | /// The length of the Ed25519Signature
```

file_excerpt · crates/aptos-crypto/src/ed25519/mod.rs · line 38 · SHA-256 fa75a35cf657ddee... · File 7b21ea56b9c4fbf3...

```
37 | /// The length of the Ed25519Signature
38 | pub const ED25519_SIGNATURE_LENGTH: usize = ed25519_dalek::SIGNATURE_LENGTH;
39 |
```

file_excerpt · crates/aptos-crypto/src/multi_ed25519.rs · line 10 · SHA-256 0d5facec0d4dac56... · File ae1c4781f7a6ba11...

```
 9 | use crate::{
10 |     ed25519::{
11 |         Ed25519PrivateKey, Ed25519PublicKey, Ed25519Signature, ED25519_PRIVATE_KEY_LENGTH,
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19662 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

36 reference(s) to X25519 across 2 file(s), e.g. crates/aptos-crypto/src/noise.rs:65. Role, judged from paths, function names and nearby code: channel key exchange (TLS, Noise, handshakes). Traffic protected by this key exchange can be recorded now and decrypted later; a hybrid exchange keeps the classical guarantees and adds post-quantum ones.

Affected asset	source-code:X25519:channel
Data source	crates/aptos-crypto/src/noise.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/aptos-crypto/src/noise.rs:65 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/aptos-crypto/src/noise.rs · line 65 · SHA-256 21f98045a9d7699e... · File 2cb8c0264ae10b0e...

```
 64 |
 65 | use crate::{hash::HashValue, hkdf::Hkdf, traits::Uniform as _, x25519, ValidCryptoMaterial};
 66 | use ring::aead::{self, Aad, LessSafeKey, UnboundKey};
```

file_excerpt · crates/aptos-crypto/src/noise.rs · line 104 · SHA-256 c42db5b8ba200f00... · File 2cb8c0264ae10b0e...

```
103 |     // e
104 |     let e_len = x25519::PUBLIC_KEY_SIZE;
105 |     // encrypted s
```

file_excerpt · crates/aptos-crypto/src/noise.rs · line 106 · SHA-256 b77110997bf970e9... · File 2cb8c0264ae10b0e...

```
105 |     // encrypted s
106 |     let enc_s_len = encrypted_len(x25519::PUBLIC_KEY_SIZE);
107 |     // encrypted payload
```

file_excerpt · types/src/network_address/mod.rs · line 8 · SHA-256 4e7aafdd34a2f6c3... · File 4975d0e2ec175a34...

```
  7 |     traits::{CryptoMaterialError, ValidCryptoMaterialStringExt},
  8 |     x25519,
  9 | };
```

file_excerpt · types/src/network_address/mod.rs · line 122 · SHA-256 74a6f19114396ccb... · File 4975d0e2ec175a34...

```
121 |     // human-readable x25519::PublicKey is lower-case hex encoded
122 |     NoiseIK(x25519::PublicKey),
123 |     // TODO(philiphayes): use actual handshake::MessagingProtocolVersion. we
```

file_excerpt · types/src/network_address/mod.rs · line 166 · SHA-256 614fe3ed0c2b75b2... · File 4975d0e2ec175a34...

```
165 |
166 |     #[error("error parsing x25519 public key: {0}")]
167 |     ParseX25519PubkeyError(#[from] CryptoMaterialError),
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols).

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19663 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (3 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

14 reference(s) to ECDSA across 3 file(s), e.g. crates/aptos-crypto/src/secp256k1_ecdsa.rs:120. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	crates/aptos-crypto/src/secp256k1_ecdsa.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC

Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/aptos-crypto/src/secp256k1_ecdsa.rs:120 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/aptos-crypto/src/secp256k1_ecdsa.rs · line 120 · SHA-256 0a1fc93a27f369c1... · File 6c3560abca72d30a...

```
119 | impl ValidCryptoMaterial for PrivateKey {
120 |     const AIP_80_PREFIX: &'static str = "secp256k1-priv-";
121 |
```

file_excerpt · crates/aptos-crypto/src/secp256k1_ecdsa.rs · line 186 · SHA-256 5fc60fc076c99261... · File 6c3560abca72d30a...

```
185 | impl ValidCryptoMaterial for PublicKey {
186 |     const AIP_80_PREFIX: &'static str = "secp256k1-pub-";
187 |
```

file_excerpt · crates/aptos-crypto/src/secp256k1_ecdsa.rs · line 289 · SHA-256 77cce12d3bdba770... · File 6c3560abca72d30a...

```
288 | impl ValidCryptoMaterial for Signature {
289 |     const AIP_80_PREFIX: &'static str = "secp256k1-sig-";
290 |
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs · line 26 · SHA-256 84af77bba23df6b8... · File 89ccb0ac1699817d...

```
 25 | #[key_name("Secp256r1EcdsaPrivateKey")]
 26 | pub struct PrivateKey(pub(crate) p256::ecdsa::SigningKey);
 27 |
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs · line 42 · SHA-256 67a910a2cba14280... · File 89ccb0ac1699817d...

```
 41 | #[key_name("Secp256r1EcdsaPublicKey")]
 42 | pub struct PublicKey(pub(crate) p256::ecdsa::VerifyingKey);
 43 |
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs · line 66 · SHA-256 77b6daa36734653a... · File 89ccb0ac1699817d...

```
 65 |     ) -> std::result::Result<PrivateKey, CryptoMaterialError> {
 66 |         match p256::ecdsa::SigningKey::from_slice(bytes) {
 67 |             Ok(p256_secret_key) => Ok(PrivateKey(p256_secret_key)),
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs · line 24 · SHA-256 a323ce9f251b5a44... · File 33d282f06a6e7d76...

```
23 | #[key_name("Secp256r1EcdsaSignature")]
24 | pub struct Signature(pub(crate) p256::ecdsa::Signature);
25 |
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs · line 41 · SHA-256 870bc135c86c31b1... · File 33d282f06a6e7d76...

```
40 |     ) -> std::result::Result<Signature, CryptoMaterialError> {
41 |         match p256::ecdsa::Signature::try_from(bytes) {
42 |             Ok(p256_signature) => Ok(Signature(p256_signature)),
```

file_excerpt · crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs · line 98 · SHA-256 3f403f5ecdb50dfc... · File 33d282f06a6e7d76...

```
97 |         let new_s_nonzero = NonZeroScalar::new(new_s).unwrap();
98 |         let new_sig = p256::ecdsa::Signature::from_scalars(r, new_s_nonzero).unwrap();
99 |         Signature(new_sig)
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19667 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BN254 (alt_bn128): transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to BN254 (alt_bn128) across 1 file(s), e.g. types/src/keyless/mod.rs:17. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:BN254 (alt_bn128):transaction_signing
Data source	types/src/keyless/mod.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open types/src/keyless/mod.rs:17 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · types/src/keyless/mod.rs · line 17 · SHA-256 cbc533ba4a0ce3f0… · File aec84ffbda933e5e…

```
16 | use aptos_crypto_derive::{BCSCryptoHash, CryptoHasher};
17 | use ark_bn254::Bn254;
18 | use ark_groth16::PreparedVerifyingKey;
```

file_excerpt · types/src/keyless/mod.rs · line 63 · SHA-256 ba7a34f492e7d3a1… · File aec84ffbda933e5e…

```
62 | /// A VK that we use often for keyless e2e tests and smoke tests.
63 | pub static VERIFICATION_KEY_FOR_TESTING: Lazy<PreparedVerifyingKey<Bn254>> =
64 |     Lazy::new(prepared_vk_for_testing);
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19661 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: role not determined (3 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

17 reference(s) to X25519 across 3 file(s), e.g. crates/aptos-crypto/src/lib.rs:34. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:X25519:general
Data source	crates/aptos-crypto/src/lib.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/aptos-crypto/src/lib.rs:34 and inspect the quoted lines; the evidence lists further
files.
```

EVIDENCE

file_excerpt · crates/aptos-crypto/src/lib.rs · line 34 · SHA-256 f134d93afe02965c... · File 6536c4a7b7bd5d55...

```
  33 | pub mod weighted_config;
  34 | pub mod x25519;
  35 |
```

file_excerpt · crates/aptos-crypto/src/x25519.rs · line 35 · SHA-256 cd421b6ecb48ccae... · File 98ce32435fbf406d...

```
  34 |     traits::{self, CryptoMaterialError, ValidCryptoMaterial, ValidCryptoMaterialStringExt},
  35 |     x25519,
  36 | };
```

file_excerpt · crates/aptos-crypto/src/x25519.rs · line 49 · SHA-256 f36c219d0438d957... · File 98ce32435fbf406d...

```
  48 | //
  49 | pub use x25519_dalek;
  50 |
```

file_excerpt · crates/aptos-crypto/src/x25519.rs · line 68 · SHA-256 62032054174ffee1... · File 98ce32435fbf406d...

```
  67 | #[cfg_attr(any(test, feature = "fuzzing"), derive(Clone))]
  68 | pub struct PrivateKey(x25519_dalek::StaticSecret);
  69 |
```

file_excerpt · types/src/account_address.rs · line 8 · SHA-256 60d89570fe4ecd22... · File ee51cce722e41a74...

```
   7 |     hash::{CryptoHasher, HashValue},
   8 |     x25519,
   9 | };
```

file_excerpt · types/src/account_address.rs · line 140 · SHA-256 a03e636a2a365185... · File ee51cce722e41a74...

```
  139 | // See this issue for potential improvements: https://github.com/aptos-labs/aptos-core/
issues/3960
  140 | pub fn from_identity_public_key(identity_public_key: x25519::PublicKey) -> AccountAddress {
  141 |     let mut array = [0u8; AccountAddress::LENGTH];
```

file_excerpt · types/src/account_address.rs · line 144 · SHA-256 60b516aca6b2d737... · File ee51cce722e41a74...

```
  143 |     // keep only the last 16 bytes
  144 |     array.copy_from_slice(&pubkey_slice[x25519::PUBLIC_KEY_SIZE -
AccountAddress::LENGTH..]);
  145 |     AccountAddress::new(array)
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19664 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (4 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

8 reference(s) to Ed25519 across 4 file(s), e.g. keyless/pepper/common/src/account_recovery_db.rs:6. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	keyless/pepper/common/src/account_recovery_db.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open keyless/pepper/common/src/account_recovery_db.rs:6 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · keyless/pepper/common/src/account_recovery_db.rs · line 6 · SHA-256 02ad39cd3ff40d72… · File 511cee839218320b…

```
5 | use aptos_crypto::compat::Sha3_256;
6 | use ed25519_dalek::Digest;
7 |
```

file_excerpt · keyless/pepper/common/src/vuf/slip_10.rs · line 7 · SHA-256 55bd92343ceed586… · File 3f246c07d30f5759…

```
 6 | pub extern crate derivation_path;
 7 | pub extern crate ed25519_dalek;
 8 |
```

file_excerpt · keyless/pepper/common/src/vuf/slip_10.rs · line 13 · SHA-256 8b5a0cd4032bfcb9… · File 3f246c07d30f5759…

```
12 | pub use derivation_path::{ChildIndex, DerivationPath};
13 | pub use ed25519_dalek::{PublicKey, SecretKey};
14 | use hmac::{Hmac, Mac};
```

file_excerpt · keyless/pepper/service/src/dedicated_handlers/handlers.rs · line 17 · SHA-256 a48683095bc2b7fa… · File 46b496f4bce4ad54…

```
16 | };
17 | use aptos_crypto::ed25519::Ed25519PublicKey;
18 | use aptos_keyless_pepper_common::{
```

file_excerpt · keyless/pepper/service/src/dedicated_handlers/handlers.rs · line 332 · SHA-256 d7e13e0ab6a536c7… · File 46b496f4bce4ad54…

```
331 |         Some(public_key_bytes) => Some(EphemeralPublicKey::ed25519(
332 |             Ed25519PublicKey::try_from(public_key_bytes.as_slice()).map_err(|error| {
333 |                 PepperServiceError::InternalError(format!(
```

file_excerpt · types/src/keyless/circuit_testcases.rs · line 20 · SHA-256 80e0e13b47f0f080… · File 792305e4068a22b3…

```
19 | };
20 | use aptos_crypto::{ed25519::Ed25519PrivateKey, PrivateKey, Uniform};
21 | use ark_bn254::Bn254;
```

file_excerpt · types/src/keyless/circuit_testcases.rs · line 151 · SHA-256 ce8a0b28f35892a1… · File 792305e4068a22b3…

```
150 |
151 | pub(crate) static SAMPLE_ESK: Lazy<Ed25519PrivateKey> =
152 |     Lazy::new(Ed25519PrivateKey::generate_for_testing);
```

file_excerpt · types/src/keyless/circuit_testcases.rs · line 152 · SHA-256 f97b86c13556a982… · File 792305e4068a22b3…

```
151 | pub(crate) static SAMPLE_ESK: Lazy<Ed25519PrivateKey> =
152 |     Lazy::new(Ed25519PrivateKey::generate_for_testing);
153 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19665 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BN254 (alt_bn128): role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

24 reference(s) to BN254 (alt_bn128) across 6 file(s), e.g. keyless/pepper/service/src/external_resources/groth16_vk.rs:6. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:BN254 (alt_bn128):general
Data source	keyless/pepper/service/src/external_resources/groth16_vk.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open keyless/pepper/service/src/external_resources/groth16_vk.rs:6 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · keyless/pepper/service/src/external_resources/groth16_vk.rs · line 6 · SHA-256 8b71194ad53a02db… · File 6b43294876f6294e…

```
  5 | use anyhow::{anyhow, Result};
  6 | use ark_bn254::Bn254;
  7 | use ark_groth16::{PreparedVerifyingKey, VerifyingKey};
```

file_excerpt · keyless/pepper/service/src/external_resources/groth16_vk.rs · line 47 · SHA-256 c92b34abfe8acfd5… · File 6b43294876f6294e…

```
 46 |     /// Converts the on-chain verification key to a prepared verifying key
 47 |     pub fn to_ark_prepared_verifying_key(&self) -> Result<PreparedVerifyingKey<Bn254>> {
 48 |         // Collect and parse the gamma_abc_g1 elements
```

file_excerpt · types/src/keyless/circuit_constants.rs · line 11 · SHA-256 8714c79b04efe1e9… · File 4e47662a9faa45b8…

```
 10 | use aptos_crypto::poseidon_bn254;
 11 | use ark_bn254::Bn254;
 12 | use ark_groth16::{PreparedVerifyingKey, VerifyingKey};
```

file_excerpt · types/src/keyless/circuit_constants.rs · line 30 · SHA-256 ac39114a4a4160ac… · File 4e47662a9faa45b8…

```
   29 | /// https://github.com/aptos-labs/devnet-groth16-keys/commit/
02e5675f46ce97f8b61a4638e7a0aaeaa4351f76
   30 | pub fn prepared_vk_for_testing() -> PreparedVerifyingKey<Bn254> {
   31 |     // Convert the projective points to affine.
```

file_excerpt · types/src/keyless/circuit_constants.rs · line 102 · SHA-256 f7d97ed84d9587a0... · File 4e47662a9faa45b8...

```
  101 | pub struct Groth16TrapdoorSetup {
  102 |     pub simulation_pk: Trapdoor<Bn254>,
  103 |     pub prepared_vk: PreparedVerifyingKey<Bn254>,
```

file_excerpt · types/src/keyless/circuit_testcases.rs · line 21 · SHA-256 b89ecf8dca334e4f... · File 792305e4068a22b3...

```
   20 | use aptos_crypto::{ed25519::Ed25519PrivateKey, PrivateKey, Uniform};
   21 | use ark_bn254::Bn254;
   22 | use ark_groth16::{PreparedVerifyingKey, VerifyingKey};
```

file_excerpt · types/src/keyless/circuit_testcases.rs · line 205 · SHA-256 7a31aaf35992b186... · File 792305e4068a22b3...

```
  204 | /// A new Groth16 VK to test the VK rotation.  Using https://raw.githubusercontent.com/
aptos-labs/devnet-groth16-keys/refs/heads/master/verification_key.json
  205 | pub(crate) static SAMPLE_UPGRADED_VK: Lazy<PreparedVerifyingKey<Bn254>> = Lazy::new(|| {
  206 |     let alpha_g1 = g1_projective_str_to_affine(
```

file_excerpt · types/src/keyless/groth16_sig.rs · line 17 · SHA-256 ac3f63a07e42fdc8... · File 9de9c6410d73e6d2...

```
   16 | use aptos_crypto_derive::{BCSCryptoHash, CryptoHasher};
   17 | use ark_bn254::{Bn254, Fr};
   18 | use ark_ff::{BigInteger, PrimeField};
```

file_excerpt · types/src/keyless/groth16_sig.rs · line 163 · SHA-256 e7018db89ae59472... · File 9de9c6410d73e6d2...

```
  162 |         public_inputs_hash: Fr,
  163 |         pvk: &PreparedVerifyingKey<Bn254>,
  164 |     ) -> anyhow::Result<()> {
```

file_excerpt · types/src/keyless/groth16_sig.rs · line 218 · SHA-256 0a7b806327385c8f... · File 9de9c6410d73e6d2...

```
  217 |         public_inputs_hash: Fr,
  218 |         pvk: &PreparedVerifyingKey<Bn254>,
  219 |     ) -> anyhow::Result<()> {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19666 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable RSA: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to RSA across 1 file(s), e.g. types/src/jwks/rsa/mod.rs:45. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:RSA:general
Data source	types/src/jwks/rsa/mod.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 16:20 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open types/src/jwks/rsa/mod.rs:45 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · types/src/jwks/rsa/mod.rs · line 45 · SHA-256 0aeed392528e804e... · File f5389d0b7d210a3c...

```
   44 |     // TODO(keyless): Hacking around the difficulty of parsing PKCS#8-encoded PEM files with
the `pem` crate
   45 |     let der = rsa::RsaPrivateKey::from_pkcs8_pem(include_str!
("insecure_test_jwk_private_key.txt"))
   46 |         .unwrap()
```

file_excerpt · types/src/jwks/rsa/mod.rs · line 62 · SHA-256 badfcb56f8b0b0bd... · File f5389d0b7d210a3c...

```
   61 |             kty: "RSA".to_string(),
   62 |             alg: "RS256".to_string(),
   63 |             e: "AQAB".to_string(),
```

file_excerpt · types/src/jwks/rsa/mod.rs · line 83 · SHA-256 d1a9a53ec8744bfb... · File f5389d0b7d210a3c...

```
   82 |             kty:"RSA".to_owned(),
   83 |             alg:"RS256".to_owned(),
   84 |             e:"AQAB".to_owned(),
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-19659	file_excerpt	crates/aptos-crypto/src/bls12381/bls12381_keys.rs (line 77)	f86b41febe322eda289a9b3744910f9f56e1842a57985dd8f18e9dfde49
QSI-2026-19659	file_excerpt	crates/aptos-crypto/src/bls12381/bls12381_keys.rs (line 80)	a99c05717793c659435d625fc91a078fa8de99cac0b7c2f81426a4891e0
QSI-2026-19659	file_excerpt	crates/aptos-crypto/src/constant_time/zkcrypto_scalar_mul.rs (line 4)	221f1dca25f14b9dfb0bda3a9eef2c447551e2ee404ab82af2906b77afb
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_keys.rs (line 9)	adf76b09b67a86ab80f62343703cd06f08aaa6ef2d74c838afbaf3087a5
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_keys.rs (line 16)	1caccba153f57a757992b88183d03a08f2b137bc1eed9afe190890307f3
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_keys.rs (line 24)	1a0100275641151db47cb652da5dffeeadab86a7c7dce2b1bbae80b7b13
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_sigs.rs (line 7)	e348dba0532f4e56f86d5b2e1db90e28010ed349c9accf0f2f0204c73a2
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_sigs.rs (line 19)	7c9df4c2d607763e63885116d7333c8f1ec08aebc1c8cec80201faf59a3
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/ed25519_sigs.rs (line 44)	2d4aff20c28e007490dbb4df38c52c7ce6cf4f507647f7fbf72494e771a
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/mod.rs (line 34)	4109af929d07df4dac5c3148e82f383831c21b4c7a72e67752f1cb2e414
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/mod.rs (line 36)	e36e845052f25c992bc097ff2643a8919a542814ca3d80a30c23bbdeefd
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/ed25519/mod.rs (line 38)	fa75a35cf657ddee121a387d195c85f31e9bd4ff176e92985131bd3a19a
QSI-2026-19660	file_excerpt	crates/aptos-crypto/src/multi_ed25519.rs (line 10)	0d5facec0d4dac56115065a9c632c1ed1a25aaa7e14a7ba15b8ac6bb992
QSI-2026-19662	file_excerpt	crates/aptos-crypto/src/noise.rs (line 65)	21f98045a9d7699e76ccf75949f7da21bb9582d0e9cce6c86a823b43280
QSI-2026-19662	file_excerpt	crates/aptos-crypto/src/noise.rs (line 104)	c42db5b8ba200f00cea1b1af6a1e59e326dfdf1720dc5852f8111f8c7b2

QSI-2026-19662	file_excerpt	crates/aptos-crypto/src/noise.rs (line 106)	b77110997bf970e95f378f3e430067fa6d2c6e9e44a6aa6d7afc327b132
QSI-2026-19662	file_excerpt	types/src/network_address/mod.rs (line 8)	4e7aafdd34a2f6c3daa63b8613f3d54065e58c42599434b49c750a80c7e
QSI-2026-19662	file_excerpt	types/src/network_address/mod.rs (line 122)	74a6f19114396ccb2c0c1d35129e8a6590a3243fe4c2ba18b831a8eabde
QSI-2026-19662	file_excerpt	types/src/network_address/mod.rs (line 166)	614fe3ed0c2b75b2f74023381798dd798583e5bc0022e8571275e815514
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256k1_ecdsa.rs (line 120)	0a1fc93a27f369c174809d52f96d68567e75108bf4af7db9ab12f333093
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256k1_ecdsa.rs (line 186)	5fc60fc076c992614ddae3137a4c2774ba1d79a69368c280f4b82969a02
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256k1_ecdsa.rs (line 289)	77cce12d3bdba7700d8f5431dce16cccfb1eb7eaf8e08e29177a821bc85
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs (line 26)	84af77bba23df6b8d90baf6a3498d3b5a95a1b4e3a08d61ffa1e111626
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs (line 42)	67a910a2cba142805ca03842c07ad5995bcd28afd9755e1dc58ec17f8d2
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs (line 66)	77b6daa36734653a5846b1caa099fbc550c118768f7bd39749cf49d52df
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs (line 24)	a323ce9f251b5a44ced790963b723501b8adabadc96888ad9a7fbfad5c0
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs (line 41)	870bc135c86c31b1a04131454f9174df611635be94fbea23c9df58f60f7
QSI-2026-19663	file_excerpt	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs (line 98)	3f403f5ecdb50dfc7676ddf953a25712ebbf6649c2f9e3f0318c7529c04
QSI-2026-19667	file_excerpt	types/src/keyless/mod.rs (line 17)	cbc533ba4a0ce3f09b8065b0bbeb02e7698f15ef6145e4fd754d42bf0ce

QSI-2026-19667	file_excerpt	types/src/keyless/mod.rs (line 63)	ba7a34f492e7d3a1c0740b7e81dac32c308ebe5fea669e29b5e31dbc995
QSI-2026-19661	file_excerpt	crates/aptos-crypto/src/lib.rs (line 34)	f134d93afe02965c942aabb6047efed4da543bcb252f8809256b8b68023
QSI-2026-19661	file_excerpt	crates/aptos-crypto/src/x25519.rs (line 35)	cd421b6ecb48ccae14cf74794f6dc5d3e6e4604dad96c9a86e09924cf38
QSI-2026-19661	file_excerpt	crates/aptos-crypto/src/x25519.rs (line 49)	f36c219d0438d9571c16a3e24335b85bdeb2d9d3f76329bf3b30f516d0a
QSI-2026-19661	file_excerpt	crates/aptos-crypto/src/x25519.rs (line 68)	62032054174ffee15e2c56a8f615a05646757644061691a29be5bedb238
QSI-2026-19661	file_excerpt	types/src/account_address.rs (line 8)	60d89570fe4ecd229fa47359b9ef69023bf2d8827a46efc259eb6796a04
QSI-2026-19661	file_excerpt	types/src/account_address.rs (line 140)	a03e636a2a3651852d60d27d8b841ba7302caf9773d571273c7cdb52be0
QSI-2026-19661	file_excerpt	types/src/account_address.rs (line 144)	60b516aca6b2d73706710a24a8e076ef7fe5272b002129931a954fa2f95
QSI-2026-19664	file_excerpt	keyless/pepper/common/src/account_recovery_db.rs (line 6)	02ad39cd3ff40d72b87d99b9b831167be159bb8a884d0ac819b09009066
QSI-2026-19664	file_excerpt	keyless/pepper/common/src/vuf/slip_10.rs (line 7)	55bd92343ceed586851337d79d69c5713464591d8dc1223743f6d2fb24e
QSI-2026-19664	file_excerpt	keyless/pepper/common/src/vuf/slip_10.rs (line 13)	8b5a0cd4032bfcb9514272fadbd1cc057daf874a5aa7482dca93265104c
QSI-2026-19664	file_excerpt	keyless/pepper/service/src/dedicated_handlers/handlers.rs (line 17)	a48683095bc2b7fa5407bd4300b9b89fbd2d2251dc96f1a3d2f45c32261
QSI-2026-19664	file_excerpt	keyless/pepper/service/src/dedicated_handlers/handlers.rs (line 332)	d7e13e0ab6a536c71db9df79afc46eb9c48297931b6a36490cd8aafa357
QSI-2026-19664	file_excerpt	types/src/keyless/circuit_testcases.rs (line 20)	80e0e13b47f0f080c33a94538f572b39340ce59d7a12ffc7c4d94c52db7
QSI-2026-19664	file_excerpt	types/src/keyless/circuit_testcases.rs (line 151)	ce8a0b28f35892a16fb066050c2e1a8d9466731b3a205679be1bbca3321
QSI-2026-19664	file_excerpt	types/src/keyless/circuit_testcases.rs (line 152)	f97b86c13556a982a6661478a0b0cb7636899a3c523c02ae57c9eadf6b0

QSI-2026-19665	file_excerpt	keyless/pepper/service/src/external_resources/groth16_vk.rs (line 6)	`8b71194ad53a02db1b15dd081670d21504c9d2bfa14c15b3003a7be1051`
QSI-2026-19665	file_excerpt	keyless/pepper/service/src/external_resources/groth16_vk.rs (line 47)	`c92b34abfe8acfd578a35eb1f6a2d59be216084165af58b9ef55a5ef65a`
QSI-2026-19665	file_excerpt	types/src/keyless/circuit_constants.rs (line 11)	`8714c79b04efe1e9428b8ac6055e6cda1f3b008f535881d839713db5609`
QSI-2026-19665	file_excerpt	types/src/keyless/circuit_constants.rs (line 30)	`ac39114a4a4160acdfcf11a9c8e9f7c7a0cc8bcea12e80685e72159e5b5`
QSI-2026-19665	file_excerpt	types/src/keyless/circuit_constants.rs (line 102)	`f7d97ed84d9587a0def5ee51627c14fd791790704b7c80a081f2504656`
QSI-2026-19665	file_excerpt	types/src/keyless/circuit_testcases.rs (line 21)	`b89ecf8dca334e4f373b14be738d126c94d3f8274b6b2b1b7dc43bf6c88`
QSI-2026-19665	file_excerpt	types/src/keyless/circuit_testcases.rs (line 205)	`7a31aaf35992b186681f27b482553c40ffd2901c7df978b53e36fe12774`
QSI-2026-19665	file_excerpt	types/src/keyless/groth16_sig.rs (line 17)	`ac3f63a07e42fdc8535a951ef251dd8fa1b4d283f116eb4f18904247b0c`
QSI-2026-19665	file_excerpt	types/src/keyless/groth16_sig.rs (line 163)	`e7018db89ae59472286666220c398aa35ed4b5d2bb6162e88b9211f8f3a`
QSI-2026-19665	file_excerpt	types/src/keyless/groth16_sig.rs (line 218)	`0a7b806327385c8f13574ddf1b99f4ba952029879a3d1feaa3757c31309`
QSI-2026-19666	file_excerpt	types/src/jwks/rsa/mod.rs (line 45)	`0aeed392528e804e81c65f28227f220c4f05ff1bdc4a98e6522f519cc10`
QSI-2026-19666	file_excerpt	types/src/jwks/rsa/mod.rs (line 62)	`badfcb56f8b0b0bdfdaca56c7b9f0fdb8ae485d74d3daa225b8e78a2fb4`
QSI-2026-19666	file_excerpt	types/src/jwks/rsa/mod.rs (line 83)	`d1a9a53ec8744bfbb8c1fda5c05f046f54da1d55ec5bf082d5e3ddeeb6a`

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 2 of 4 — post-quantum in use outside the critical paths

- In use: SLH-DSA — types/src/on_chain_config/aptos_features.rs
- Present, not wired: SLH-DSA — crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_keys.rs
- Present, not wired: SLH-DSA — crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_sigs.rs
- A post-quantum library is declared as a dependency
- Classical only — channel key exchange (TLS, Noise, handshakes): X25519 (2 file(s))

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
channel key exchange (TLS, Noise, handshakes)	X25519 (2)	QSI-2026-19662	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
transaction and message signing	Ed25519 (10), ECDSA (3), BLS12-381 (2), BN254 (alt_bn128) (1)	QSI-2026-19659, QSI-2026-19660, QSI-2026-19663, QSI-2026-19667	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them

role not determined	BN254 (alt_bn128) (6), Ed25519 (4), X25519 (3), RSA (1)	QSI-2026-19661, QSI-2026-19664, QSI-2026-19665, QSI-2026-19666	Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	8
Grover: acceptable to NIST, below ECCG/ANSSI guidance	12
n/a	14
post-quantum	4
vulnerable (Shor)	58

Detailed findings: QSI-2026-19659, QSI-2026-19660, QSI-2026-19662, QSI-2026-19663, QSI-2026-19667, QSI-2026-19661, QSI-2026-19664, QSI-2026-19665, QSI-2026-19666

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (2)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Shor's algorithm — discrete logarithms in finite fields Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (6), BN254 (alt_bn128) (8), ElGamal (1), Groth16 (3)	Breaks finite-field Diffie–Hellman and DSA at every size. Move key establishment to ML-KEM (hybrid with X25519 or ECDH during the transition) and signatures to ML-DSA or SLH-DSA.

Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (6), BN254 (alt_bn128) (8), ECDSA (6), Ed25519 (24), Groth16 (3), X25519 (8)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	AES-256 (2), HKDF (2), HMAC (2), Poseidon (9), SHA-2 (1), SHA-3 (4), SLH-DSA (3)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Poseidon (9), SHA-2 (1), SHA-3 (4)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.
Simon's algorithm — superposition attacks on symmetric constructions Q2: superposition queries to the victim's keyed primitive	only with superposition queries	GCM mode (1)	No action for ordinary deployments of AES-GCM or CBC-MAC — the attack model does not apply. Record it as a model assumption; see the offline variant for Even–Mansour-type designs.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- osv_advisories — not checked: no exactly pinned versions to query

Remediation plan

QSI-2026-19659 · P3 · 90 days

short-term risk reduction

Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19660 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19662 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19663 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19667 · P3 · 90 days

short-term risk reduction

Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19661 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19664 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19665 · P3 · 90 days

short-term risk reduction

Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19666 · P3 · 90 days

short-term risk reduction

Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19659 · P4 · 90 days

root-cause fix

Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-19660 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19662 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-19663 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19667 · P4 · 90 days

root-cause fix

Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-19661 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-19664 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19665 · P4 · 90 days root-cause fix

Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-19666 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.

Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-19659	Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19660	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19662	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19663	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19667	Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19661	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19664	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19665	Add BN254 (alt_bn128) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19666	Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-19659	Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-19660	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19662	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19663	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19667	Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19661	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19664	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19665	Plan migration of BN254 (alt_bn128) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19666	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-19659	Re-run the inventory after refactoring.
QSI-2026-19660	Re-run the inventory after refactoring.
QSI-2026-19662	Re-run the inventory after refactoring.
QSI-2026-19663	Re-run the inventory after refactoring.
QSI-2026-19667	Re-run the inventory after refactoring.
QSI-2026-19661	Re-run the inventory after refactoring.
QSI-2026-19664	Re-run the inventory after refactoring.
QSI-2026-19665	Re-run the inventory after refactoring.
QSI-2026-19666	Re-run the inventory after refactoring.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
AES-256	AES-256	reduced margin (Grover), approved	acceptable	2	2	crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs:15; crates/aptos-crypto/src/noise.rs:126
HKDF		reduced margin (Grover), approved	acceptable	2	2	crates/aptos-crypto/src/hkdf.rs:116; crates/aptos-crypto/src/noise.rs:65
HMAC		reduced margin (Grover), approved	acceptable	1	1	keyless/pepper/common/src/vuf/slip_10.rs:14
HMAC	HMAC-SHA-512	reduced margin (Grover), approved	acceptable	1	1	keyless/pepper/common/src/vuf/slip_10.rs:55
SHA-3	SHA3-512	reduced margin (Grover), approved	acceptable	2	2	crates/aptos-crypto/src/arkworks/hashing.rs:41; keyless/pepper/service/src/vuf_keypair.rs:73
Poseidon		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	9	9	crates/aptos-crypto/src/lib.rs:24; crates/aptos-crypto/src/poseidon_bn254/constants.rs:4; crates/aptos-crypto/src/poseidon_bn254/keyless.rs:6
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	1	1	crates/aptos-crypto/src/noise.rs:189
SHA-3	SHA3-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	2	2	crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs:32; keyless/pepper/common/src/account_recovery_db.rs:28

Crypto library ed25519-dalek (ed25519-dalek)	workspace	n/a	acceptable	2	2	crates/aptos-crypto/Cargo.toml; keyless/pepper/common/Cargo.toml
Crypto library x25519-dalek (ed25519-dalek)	workspace	n/a	acceptable	1	1	crates/aptos-crypto/Cargo.toml
Crypto library bls12_381 (bls12_381)	workspace	n/a	acceptable	1	1	crates/aptos-crypto/Cargo.toml
Crypto library curve25519-dalek (ed25519-dalek)	workspace	n/a	acceptable	1	1	crates/aptos-crypto/Cargo.toml
Crypto library ring (ring (Rust))	workspace	n/a	acceptable	2	2	crates/aptos-crypto/Cargo.toml; types/Cargo.toml
Crypto library p256 (RustCrypto elliptic curves)	workspace	n/a	acceptable	2	2	crates/aptos-crypto/Cargo.toml; types/Cargo.toml
Crypto library jsonwebtoken (jsonwebtoken (Node))	workspace	n/a	acceptable	3	3	keyless/pepper/common/Cargo.toml; keyless/pepper/service/Cargo.toml; types/Cargo.toml
Crypto library rsa (RustCrypto rsa)	workspace	n/a	acceptable	1	1	types/Cargo.toml
GCM mode		n/a	acceptable	1	1	crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs
PQC library slh-dsa (RustCrypto post-quantum crates)	workspace	post-quantum	acceptable	1	1	crates/aptos-crypto/Cargo.toml

SLH-DSA		post-quantum	acceptable	3	3	crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_keys.rs:18; crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_sigs.rs:16; types/src/on_chain_config/aptos_features.rs:170
BLS12-381		vulnerable (Shor)	acceptable	6	6	crates/aptos-crypto/src/bls12381/bls12381_keys.rs:77; crates/aptos-crypto/src/bls12381/bls12381_pop.rs; crates/aptos-crypto/src/bls12381/bls12381_sigs.rs
BN254 (alt_bn128)		vulnerable (Shor)	acceptable	8	8	keyless/pepper/service/src/external_resources/groth16_vk.rs:6; types/src/keyless/bn254_circom.rs; types/src/keyless/circuit_constants.rs:11
ECDSA	secp256k1	vulnerable (Shor)	acceptable	1	1	crates/aptos-crypto/src/secp256k1_ecdsa.rs:120
ECDSA		vulnerable (Shor)	acceptable	3	3	crates/aptos-crypto/src/secp256r1_ecdsa/mod.rs:44; crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs:26; crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs:24
ECDSA	P-256	vulnerable (Shor)	acceptable	2	2	crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs:26; crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs:24
Ed25519	Ed25519	vulnerable (Shor)	acceptable	24	24	crates/aptos-crypto/src/ed25519/ed25519_keys.rs:16; crates/aptos-crypto/src/ed25519/ed25519_keys.rs:9; crates/aptos-crypto/src/ed25519/ed25519_sigs.rs:7
ElGamal		vulnerable (Shor)	deprecated	1	1	crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs
Groth16		vulnerable (Shor)	acceptable	3	3	keyless/pepper/service/src/external_resources/groth16_vk.rs; types/src/keyless/groth16_sig.rs; types/src/keyless/groth16_vk.rs
RSA		vulnerable (Shor)	acceptable	1	1	types/src/jwks/rsa/mod.rs:45
RSA	RS256	vulnerable (Shor)	acceptable	1	1	types/src/jwks/rsa/mod.rs:62
X25519		vulnerable (Shor)	acceptable	2	2	crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs; crates/aptos-crypto/src/elgamal/curve25519.rs

X25519	X25519	vulnerable (Shor)	acceptable	6	6	crates/aptos-crypto/src/lib.rs:34; crates/aptos-crypto/src/noise.rs:65; crates/aptos-crypto/src/x25519.rs:35

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 496bbf2c0ceaa6407d995b8152ee8fd44f8f95ed1f641fbd08068a5bb313751f

C. Input files

File	Size	SHA-256	R
aptos-move/framework/src/aptos.rs	8542	fcf083dd26c8621f3790d2135561197bb2dde15dcb4dbc10281080bac9417997	0
aptos-move/framework/src/built_package.rs	28767	c35c94adfde8f8bae53b6a85aecd0cf23c026c502a7f277eff372911e743d1a8	0
aptos-move/framework/src/chunked_publish.rs	7922	0147d227083a1a99f03dc98e5b7d4902d782bdccc7278c7cfadfe0eb28926888	0
aptos-move/framework/src/docgen.rs	5267	998a21a6a0e5e57e6717808d00f66fc4ed0cca1c25601881b6c9a35cf5f2386b	0
aptos-move/framework/src/extended_checks.rs	38284	bf18b4a2fb461aaa4810fd4f457c43a8a77114eafb66e7c6d562acbc5a99806a	0
aptos-move/framework/src/lib.rs	1308	bf55073178dacf1fa2f2201687a8d730604cbb0fbb4a76ccdfa55a5ac0aaaaf8	0
aptos-move/framework/src/main.rs	2975	79e70277296b372102d4b7db1a5252fc29731b0e01f6b95406f6a91ecafa4197	0
aptos-move/framework/src/prover.rs	19222	36d2b5eb124c63e6819579c0a20e4ac4adda69bdb9198c6c1214f58046d0be9e	0
aptos-move/framework/src/release_builder.rs	5125	d3b433b4191c96586f92ce2e8443a3f550502eae0f1fb02246b55210479e3b1e	0
aptos-move/framework/src/release_bundle.rs	1428	650f244c4402a6bd7cc5f00d93f62d3cc35a24fd1be68f872ab86ae8dc9bb658	0
crates/aptos-crypto/Cargo.toml	4150	427240ded930807125514251be202b173d814c54dbfe1196963c5d5bddfe895b	0
crates/aptos-crypto/src/arkworks/differentiate.rs	2641	535935d73d2ee8b9052c0df97a5c13b295a4e61b002538af747176d8dc599222	0
crates/aptos-crypto/src/arkworks/hashing.rs	4378	f6423e811efd0e94961d3d799f1dc6d2a82985a44793a3f14c5f2a8a78b51675	0
crates/aptos-crypto/src/arkworks/mod.rs	4122	a169eeb3e3a5c14c46b75b1698ccee9933e4df801c361f51a4e747ce12e3dd07	0
crates/aptos-crypto/src/arkworks/msm.rs	4829	c37af58ba43dd67dd988086d66a2f2676d0b8f3d11f8b9d8788c722cf73d0345	0
crates/aptos-crypto/src/arkworks/random.rs	4979	c1c21ed11f1faaff52c99a588b61dbafd74aaf2b84fd31a9f05476fcd162c186	0
crates/aptos-crypto/src/arkworks/scrape.rs	11248	d5a08178f5a24d15c34c9765486cf472cd1c5592c81407d5a470ab521b99db9d	0

crates/aptos-crypto/src/arkworks/serialization.rs	4958	f2df1c7f19a0aac6d4809f62a1e558ccebf14f119dbce3762cb2617693a3241f	0
crates/aptos-crypto/src/arkworks/shamir.rs	18514	4739f5e1c70ec6cf2475cbbd711d19f992dbbb4035cd812f84794f0e803ebae5	0
crates/aptos-crypto/src/arkworks/srs.rs	8411	0e11689ff310df143a22d526923c88259149b1f7c88b32d7fc810cf35701c8af	0
crates/aptos-crypto/src/arkworks/vanishing_poly.rs	3321	b2645ccc303cf0cadf6a5eb7ab59c61ef77d638eb78c46ce382a402907899fdb	0
crates/aptos-crypto/src/arkworks/weighted_sum.rs	1365	702231ffd20ad9eb6a3a7d181b22983fce1d4eddca150ea830bd1c973449b942	0
crates/aptos-crypto/src/asymmetric_encryption/elgamal_curve25519_aes256_gcm.rs	6045	6b01dda952ebe1960565b297a2d67fac4572c53baeecde311dbb24ca90f740a1	0
crates/aptos-crypto/src/asymmetric_encryption/mod.rs	1491	4f58c0c94e7c521cc7455d2c25290e35dad6b0e523278b22b56542c91c77cc4d	0
crates/aptos-crypto/src/bls12381/bls12381_keys.rs	10760	b2ebf8767ada59dc70cc0b6b01f525d9bdcb77753bd527770db30b44f476e512	0
crates/aptos-crypto/src/bls12381/bls12381_pop.rs	6512	ce93253d1294e232768f51235d4d29145f569e38a9f97df18de89d13405a5c38	0
crates/aptos-crypto/src/bls12381/bls12381_sigs.rs	8454	76831485928ebc9dd3d96224d913361f7a120b60f8e714bad1bf6856735843b8	0
crates/aptos-crypto/src/bls12381/bls12381_validatable.rs	7303	d17948f1d6363296bd855204618c52428ec0730f78d1bf0c49b8193361cab57e	0
crates/aptos-crypto/src/bls12381/mod.rs	23250	4afd88a46f7d49ae01d78c704e3e7b1d09897992133e72f943cdff82c09d1311	0
crates/aptos-crypto/src/blstrs/evaluation_domain.rs	8024	9c9a0eb6de846e719b24297ec68b7bc833ea19189aeb93d2c708da870d05712f	0
crates/aptos-crypto/src/blstrs/fft.rs	6631	76306cde79adc8f4610ee2e77f61b12f0f48c185a645b9452ff255673d883a91	0
crates/aptos-crypto/src/blstrs/lagrange.rs	15244	7c88a82115ad016d6f8ab4f2f5eae35018ee181c8fd873cbe5f23b3aa9ab6f47	0
crates/aptos-crypto/src/blstrs/mod.rs	6860	95fde1ecccf049114d882eb20d14570ec73505fa02e2d26512440062514cf2bd	0
crates/aptos-crypto/src/blstrs/polynomials.rs	21122	79759c144e0ad0bcda97805d0236024f94833f92f5743a90879a93166bc9626e	0
crates/aptos-crypto/src/blstrs/random.rs	5920	77ed170dbaadf8deec686583675f69c76ce766a8470e86c4ed5a922c1d01e29a	0
crates/aptos-crypto/src/blstrs/scalar_secret_key.rs	1492	98334ee57e6a96bf2ead9f634e86d57285437400c853c0b3d01bbce3523be3d7	0
crates/aptos-crypto/src/blstrs/threshold_config.rs	4858	342cda1d07632e5614f8782940dd49654eed26a4ea0f94c2e7911fee6dd085a5	0
crates/aptos-crypto/src/bulletproofs/mod.rs	425	d846f3b90cc120939fb006a8c5913e0f4f9d339eaa631ca58a0fbeef0bb456bf	0

crates/aptos-crypto/src/compat.rs	1705	c18f5c5ad97ebf51b82c2be6e3ac016b1c0f774d97ef34900301a84c0d31fe7e	0
crates/aptos-crypto/src/constant_time/blstrs_scalar_mul.rs	4472	afa62e1a6d411f2a4ec0f8f18398aefe3aa70482e5c49884085e1445fefcac0e	0
crates/aptos-crypto/src/constant_time/mod.rs	537	fa46c51086a77b79a861fd4ea8dda3b48c691c97d87ed3fe81e0d82f4e29eb44	0
crates/aptos-crypto/src/constant_time/zkcrypto_scalar_mul.rs	3387	c791efc46427451263e753218ac6a2b1a74c9192762c79a971d5c9fdcb0f56b0	0
crates/aptos-crypto/src/ed25519/ed25519_keys.rs	12368	79a6f426fb5efb58ae8e072947d4e9347eda7cc281509f2dff57263559b8c4a9	0
crates/aptos-crypto/src/ed25519/ed25519_sigs.rs	7998	838e7e95fe58782ee363ebf441d73b32f13e4987712a4e23c8c6b7d45b281856	0
crates/aptos-crypto/src/ed25519/mod.rs	2187	7b21ea56b9c4fbf31f19ae1789d1bfe612ffa33d990560cb65a8b2f958c38d58	0
crates/aptos-crypto/src/elgamal/curve25519.rs	1196	6d54354f3f24d7a6e7bc76b20ae3b52d3cc6443d7ab2d7616cc1f150c221ec51	0
crates/aptos-crypto/src/elgamal/mod.rs	2500	83d8912dfc5e3d2deac79735478945fcbdea9918ef912e5ddec4fab3c79f5bfd	0
crates/aptos-crypto/src/encoding_type.rs	4238	db4dcf8c76d0bc883ad6bab893154ea447fdebf9bb28a1443367d95fdb8a74b6	0
crates/aptos-crypto/src/hash.rs	29640	5e92ab53b41b8b98cf24866c26693e970131cff99c48f01f7329a700f608f500	0
crates/aptos-crypto/src/hkdf.rs	9334	c12e0c2a3ced207bdeb0f4fcc649f81ec011e96d7b1754f6e129174b903d6883	0
crates/aptos-crypto/src/input_secret.rs	2336	ebf3023280af59784f347d07b7d14de79ec2617f7ace22781d99dedcf48baa8b	0
crates/aptos-crypto/src/lib.rs	1064	6536c4a7b7bd5d55ddc372edf35c88335b09d18de020015aebb4c482143dd907	0
crates/aptos-crypto/src/multi_ed25519.rs	23396	ae1c4781f7a6ba11430ff69a738a66c695301f1366134b1102b69b4f9657ceae	0
crates/aptos-crypto/src/noise.rs	24482	2cb8c0264ae10b0e55f89da85b7505bce380a505886e62d210a86a0524968c3e	0
crates/aptos-crypto/src/player.rs	1098	a762b8ffb60ea539a33d8eb9161fa6e7d4dbe14fa2972f7f5eac75904f33e2c8	0
crates/aptos-crypto/src/poseidon_bn254/alt_fr.rs	1187	bdf936df121cc626c36266d86160148c9dcbabd1812a42c21c62029bff5afef1	0
crates/aptos-crypto/src/poseidon_bn254/constants.rs	1311506	77b60d61b5d52f231b1e481ab1999b4f5210e808aceccac75ab083945376dd1a	0
crates/aptos-crypto/src/poseidon_bn254/keyless.rs	14848	7e842fd4c814c22c11d11f99432389e8826e5db1dd46d1de8ed56e6dd14a06a5	0
crates/aptos-crypto/src/poseidon_bn254/mod.rs	5914	e8c1710988df9cfd67e60cc353c72f0e289a226ffc521b955b8a695f2b2a076a	0
crates/aptos-crypto/src/secp256k1_ecdsa.rs	9137	6c3560abca72d30a926e776c6d8575440ff115c2b2e95f90818b04079a2b305d	0
crates/aptos-crypto/src/secp256r1_ecdsa/mod.rs	2839	28d55d895077405cd4eb4c741b4ea329cb9cdd04097156d5e804e568f174e5a7	0

crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_keys.rs	9340	89ccb0ac1699817da1b5220747870eaa01ba06f2356e33f8d75a7e85cf16f719	0
crates/aptos-crypto/src/secp256r1_ecdsa/secp256r1_ecdsa_sigs.rs	7381	33d282f06a6e7d76d77dd13cbe304689838f5dd2ca56007948dc39418833317c	0
crates/aptos-crypto/src/slh_dsa_sha2_128s/mod.rs	2115	b13baf6f2dde07bd7ba865af2c1d456f7094675cb54a71295d089fa02e64515f	0
crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_keys.rs	18824	24fec169f9f85b7bc650011fa43b338ec2f7629640083c5ee600541a02065c9e	0
crates/aptos-crypto/src/slh_dsa_sha2_128s/slh_dsa_sigs.rs	4351	0547e0962260a4b1dbd297f96b21ed69858d694ac2504e58bacc880aa3973c82	0
crates/aptos-crypto/src/traits/mod.rs	15507	eaf0b9771e5ed0798af939e0a7895ec215bf5393993d49dbfb40ae4d03d0becf	0
crates/aptos-crypto/src/utils.rs	1593	c1c20941082db1db8145b46819df485ea5ae61ba06c4cdb5481a0f86e5854255	0
crates/aptos-crypto/src/validatable.rs	4700	3bc1a50fa2466b6e07aaa454fb71fae3083eb3cbdd77cd0d4a1339be47ecf881	0
crates/aptos-crypto/src/weighted_config.rs	21569	bd5e96b85159ca83d4eee684db0bb3a49657111f62903331b10cacc2ffe5139a	0
crates/aptos-crypto/src/x25519.rs	9254	98ce32435fbf406d0b5f541260fefdb82a3fbbf7a734fec59d221fe79e23245c	2
keyless/pepper/common/Cargo.toml	1035	4bf6025c80b280d8415c369feb7c36171037890a69c6da51ca301e286beec498	0
keyless/pepper/common/src/account_recovery_db.rs	1455	511cee839218320b0c80a0d6d39a88beed134b4f64d71adf0ab40de26e7c7f18	0
keyless/pepper/common/src/jwt.rs	1112	bfb8232ced840395778af71de3001b2a8454ff5062ee6e21239980550c72fd73	0
keyless/pepper/common/src/lib.rs	7312	b594ae7bb2cd19a292af6145791a7ee0bbf781ca040116195fa692e232ce584d	0
keyless/pepper/common/src/vuf/bls12381_g1_bls.rs	4184	bb3439691e1861fae253498ff947dd5285630382a07786a7d4850a5ade1a7caa	0
keyless/pepper/common/src/vuf/mod.rs	1044	3d8f6ae49835b02a879e83fd4a0c4160a6d3978a1d4163f193fac5ec27041fa0	0
keyless/pepper/common/src/vuf/slip_10.rs	6293	3f246c07d30f5759f0eec0c7600a2cd892e075aeb41c05964049e308c6ee39bc	0
keyless/pepper/service/Cargo.toml	1658	c1b332270bb1fdd5d25aaeb76ba6df4a593e929f404426903197212ee7282f90	0
keyless/pepper/service/src/accounts/account_managers.rs	3653	05ab7709bf7b451e8ace13a122b95594968119a60509cb1dcbe9f989a79778ba	0
keyless/pepper/service/src/accounts/account_recovery_db.rs	9098	c91f59771ac154d3c2160f1b6784155dbf37d2481ced46485baf06bd468a1ee9	0
keyless/pepper/service/src/accounts/mod.rs	231	332701b0f73fb4d8f9b1a1206a16be259154da921420e189446600741064d934	0
keyless/pepper/service/src/dedicated_handlers/handlers.rs	17198	46b496f4bce4ad545fc06c386c0e11adf0630d00069866a8ed1da3d913577522	0
keyless/pepper/service/src/dedicated_handlers/mod.rs	218	bd485aa22d27e24e7fc19d7a82033f47b27a114ea8bc0fbd4bd30198d32e373a	0

keyless/pepper/service/src/dedicated_handlers/pepper_request.rs	28733	f50bbf77d32067c6a7a4d21222f0125adf3367eed90a595515da14edba7c9aab	0
keyless/pepper/service/src/deployment_information.rs	1339	e8b2c33837785a6d4f62455adccb35140a18faeb86aa97a6872d6b4a26f892ea	0
keyless/pepper/service/src/error.rs	832	378a3aa9dabebd46127dd4dd03c22b8d00395aab8ee3463591a6720422a536df	0
keyless/pepper/service/src/external_resources/groth16_vk.rs	7570	6b43294876f6294ecfa0953d3d7f4ef952ba458c0f36554a9e279e38d05961b0	0
keyless/pepper/service/src/external_resources/jwk_fetcher.rs	11533	66e6d3b59794b557e853ebb7227c0d441b33752fdcd125d063f26938e9cdbf34	0
keyless/pepper/service/src/external_resources/jwk_types.rs	5420	7d9a8f275c836dbca5cfeee5da2a9839be1a93c48a96824a6395495c3e67e499	0
keyless/pepper/service/src/external_resources/keyless_config.rs	5074	b4b12f28129292891dfd565ecfc2635e4e4a0c0450d8bdd8c88ce4529dd284c9	0
keyless/pepper/service/src/external_resources/mod.rs	286	30c753f788d2223c5436d8457320db0bf380919daa1275205ffac703d9b743bf	0
keyless/pepper/service/src/external_resources/resource_fetcher.rs	10624	99e0bba35f52c92f9ff8fe59a258dd402581664ff48512781306acf9f2f24965	0
keyless/pepper/service/src/lib.rs	400	822c5b5d84aa0c374f18b4d704b43b260eaa6afa8914a7fe295b96db1f38cb6d	0
keyless/pepper/service/src/main.rs	22130	f17845f1a77d0fc059718b199fed388c5806801dca7ddbc1daf0038e8f0aee73	0
keyless/pepper/service/src/metrics.rs	5820	4ce9ee9beab799c1ffc8a1605baaa3121cdb924f1e979a4cc670390bbc4284f4	0
keyless/pepper/service/src/request_handler.rs	16211	bc5f4ac466b5ef451296854e8d7235ec92a6cf6358dd520ff2b073e4bcbb3d7b	0
keyless/pepper/service/src/utils.rs	4725	f71a94fe71c1d7e586e2d0e9eafc696229fd9be28117fe45563070311f50a65f	0
keyless/pepper/service/src/vuf_keypair.rs	9269	6fff64509d9751551aa54b9eb761c706248cf282110c0b22e508e850edde694f	0
types/Cargo.toml	3901	4b2f8b535413b475490f0f28c5e914d704ae7acc2e212da99dfaec951952a2bd	0
types/src/access_path.rs	8030	ecf1c87c4ae66b9d015a3a79e461d30b7c6053424feea1d266f60621531a11e6	0
types/src/account_address.rs	9982	ee51cce722e41a743f94cf1fd4d0a38034ff59bf9fb2682700437155f1a134bb	0
types/src/account_config/constants/account.rs	291	66b83be874fa71d9e9e169766ccdf54e5f7ff0ae4143aeb1c4cf8577834478cf	0
types/src/account_config/constants/addresses.rs	591	e75ffbffa7ede1a76c512107626319667e0c717af818a00459b6048e54f6ad3b	0
types/src/account_config/constants/mod.rs	255	9bd540838b6c6a1416e55c53df5c70cf2788def18312b7d9c9e338c1f97be5b3	0
types/src/account_config/events/burn.rs	1892	c24924e6e051c9a5a6f42f992a0dc1675262e8754a203ec12c0dbffeabfffa88	0
types/src/account_config/events/burn_event.rs	1504	70fb45d7c905e922484a965a63d7aa603b4f03934a00a732fb3ff7edc245a819	0

types/src/account_config/events/burn_token.rs	1636	0a2147dd902675a676161525e3a29c72a3a3989fa7b1d30c32b843aa1a5a2716	0
types/src/account_config/events/burn_token_event.rs	1461	2c9311164a2eb5533132ac625a0f86ccc1e8db9253a0d67c8203a62478e5ac8e	0
types/src/account_config/events/cancel_offer.rs	1915	26ffa2a4458524e28ee720975f3d998f4d8088277a02a93a733743d1d1ddfb16	0
types/src/account_config/events/claim.rs	1872	9c50ef85c7c28b526c5a97799802976c4066c0032918557bcc35ed8e58d50067	0
types/src/account_config/events/coin_deposit.rs	1690	16757d28a363d279c916ad9a4cac1d686e93e6789f9c1347321f335b503e59fc	0
types/src/account_config/events/coin_register.rs	1492	6c5292bbdf3064dda11193ba272e910a8634adef6ecaaef3ab80b5f6005c3a8f	0
types/src/account_config/events/coin_register_event.rs	1384	4c3e53fa0ad000b3924bbd74ee6a949149b139d770e769073cf67e8893fe622b	0
types/src/account_config/events/coin_withdraw.rs	1697	d735f625899b5e8a34716ef5565975b1c8ea4a660759ccafbe7931b872e865a9	0
types/src/account_config/events/collection_description_mutate.rs	2119	df20e4d5d529b09805b71073f3cd0ed0bb6b5aa4fb518ea9e9aafe77ed5a6c37	0
types/src/account_config/events/collection_description_mutate_event.rs	2155	82dd47cb0882373b42989c339e4936b2b7e782e462810bedcf62d670a60f27e7	0
types/src/account_config/events/collection_maximum_mutate.rs	2033	e4c42e3033f660b03a79ee7dceef7bd2172cb8c86353c1a5f5afa6b3fa22c974	0
types/src/account_config/events/collection_maximum_mutate_event.rs	2185	c69f706fa4db31fb48f909182b11b9320aa9c2b938e23721cef320d9aae8bffd	0
types/src/account_config/events/collection_mutation.rs	1957	c7a1861aaffe4baa5ce79a8b7da397aa6eaae9dbfab506a8eb9ce22bbad013f4	0
types/src/account_config/events/collection_mutation_event.rs	1491	5563136b89ee2088893d80fdc3c3c17fafb8fc5292b13156b5f37c0d9e2cc204	0
types/src/account_config/events/collection_uri_mutate.rs	1983	66001f77360aa4a77f2cfee42c0d08487399434c68870de2762a75076bebac26	0
types/src/account_config/events/collection_uri_mutate_event.rs	2019	30eb9279685e2b2690b761ba68bebb1b189aa3e147e4448c66ae1583822b76e7	0
types/src/account_config/events/create_collection.rs	2036	eb758dd19135a11c2ebfb1880c364180db1dfe0c34d8920a3066733d2b41c0a4	0
types/src/account_config/events/create_collection_event.rs	2072	13cc4d8c0de81603dd1bf8f024e226338cd4a4d27418f18d679bc8d70e417d3b	0
types/src/account_config/events/create_token_data_event.rs	3489	22c9e6ff7cfe18dd4869de248c0f8f117273e9dd31cd59eb535bae812525e46a	0
types/src/account_config/events/default_property_mutate.rs	2555	329a5821be637f5b514ea9db6a449cade3213cdb4da73280b51de7f84d2c390c	0

types/src/account_config/events/default_property_mutate_event.rs	2439	5af56b944274f5a93f0ae3b138dd3862cdb1305919f93702f788213fd58c4d09	0
types/src/account_config/events/deposit_event.rs	1372	d4f4ad058d40936f68a1f05ac171ee626ef132764632909f9a063155f309468b	0
types/src/account_config/events/description_mutate.rs	2123	354ea5d42da1e9cef90dffa5e2f8370b98dedbfa51580a7bfb5ec254abd9887e	0
types/src/account_config/events/description_mutate_event.rs	2159	04da68f4566a7e30da91c1acd202b714ebe15acead81dffb9198cf447afa71d9	0
types/src/account_config/events/fungible_asset.rs	1190	d71f8a7d4c4c95e78d996e133115b4dd9dbdf85bb6c156480ed6b9ed94b51d25	0
types/src/account_config/events/key_rotation.rs	1598	b02613ba7e8ee7c2d8b8c1cac22530cf2acf5176ab7093cffd13323c58dff59e	0
types/src/account_config/events/key_rotation_event.rs	1687	2af804a0ee759c1d88b94b670fbe614331a481877580e816f52211fa79ea4b21	0
types/src/account_config/events/maximum_mutate.rs	2037	2c7bb224f820f453295a3dd6dc49d45b966d9b17f12e9de6e218f42789ff525a	0
types/src/account_config/events/maximum_mutate_event.rs	2189	ccb5198dda28140413a3946ddb1115bb203ecfa9fd1c00adf2993bfa3a559cf3	0
types/src/account_config/events/mint.rs	1829	1903501a2b45d7fd7af8e0bacad5f00c6e80cef5557b2d6c00b2e62ff7cc8857	0
types/src/account_config/events/mint_event.rs	1482	6a4e3c1cec112ad8344e54c88e98c2deb54c80937891e6aa0009ad897c2b508f	0
types/src/account_config/events/mint_token.rs	1652	b92411631c4f4677a104c7cc0c6cdc715e75529a652a9885a03b5e005542bd0e	0
types/src/account_config/events/mint_token_event.rs	1477	ffedff6de8b976b72cddba5c524223d77e6d53569835c3f0b2abf69711e6e311	0
types/src/account_config/events/mod.rs	3768	cae46428243ccb9183462728b1a88f3624d2ac530e0e493766c4b3730b4b66d7	0
types/src/account_config/events/mutate_property_map.rs	2191	01bf08c95d4f778c828f962006b6e532bc72149ff543b2635b89f786f8df4e59	0
types/src/account_config/events/mutate_token_property_map_event.rs	2067	089ef272e40016c15dbff6e31b7bb49eb2eef0a5055236f9e7dd962a87ae1abc	0
types/src/account_config/events/new_block.rs	3422	6c51af4396767ba1e8e27a15fa54dff11a5aca7df789a1c4c2f17fad35cdb74e	0
types/src/account_config/events/new_epoch.rs	1440	e32383a2a078bb6345d0972c56b0bde5b9b62f16749444b16f08d35cd7d2d37e	0
types/src/account_config/events/offer.rs	1872	75519848e72df15801c2f7e968fdb1a48d1a7c9fbc90d19dd9bd2f2fe53c9814	0
types/src/account_config/events/opt_in_transfer.rs	1612	b38bea4b0c23e8cb41f7db331594c56dad898a58f743318aaf6c9608184e19d5	0

types/src/account_config/events/opt_in_transfer_event.rs	1399	5963affa223809b4689bd48dd9c1ee85c8fa3e6f0ec1e8191d013ad9a701d113	0
types/src/account_config/events/randomness_event.rs	452	3974575a5db8370988527fc4ccf8bbf142464f9b22bddfa5e427e11db46a05a7	0
types/src/account_config/events/royalty_mutate.rs	3017	233369e803809a257377f6474981eb89344daaa97e419752f6b785cf8ff362a4	0
types/src/account_config/events/royalty_mutate_event.rs	3053	829f203adcac2673f54aadf8793b3523a36bff5b279982b5a33f707a94748d21	0
types/src/account_config/events/token_cancel_offer_event.rs	1797	02034248d5773d0ab8ef36c121df52f4acece99e6af945855c9ab3de82c7fc6e	0
types/src/account_config/events/token_claim_event.rs	1754	7374dbd5f68ff8bbb1a15b53ed0b1ea4e246cb03cec7cfcfcd3048901520d02d	0
types/src/account_config/events/token_data_creation.rs	3773	9b854a1a1b484ef05f42858d95640fed0f103105b04c88a52413bde4c1f37083	0
types/src/account_config/events/token_deposit.rs	1918	2ab3a3dc953576c5d92d07f5730317e31e50c2966800980790106184b8c8fdb0	0
types/src/account_config/events/token_deposit_event.rs	1472	5158edcc6a3527bfd8897e2c4e21b92657e2ea8d3ccf393eb97c967645f00c0b	0
types/src/account_config/events/token_mutation.rs	1972	a31ff06bde4633c2d137dd9b6fd2b8ac7c17b589941f449053ab6d18e91f0d63	0
types/src/account_config/events/token_mutation_event.rs	1751	f9209e94a9b5826881bf793cf12cea919292e6cab98397d380a70f96dc7f9845	0
types/src/account_config/events/token_offer_event.rs	1754	68c6306735f87252bd206ec16b9c76c9fef8ec70d3926606731480dee2ccd404	0
types/src/account_config/events/token_withdraw.rs	1674	3fcc76d0c18d28be43f919ea32643006f2fc9442b0fd758fe286f9c2747d11c8	0
types/src/account_config/events/token_withdraw_event.rs	1479	75a5b0c64f591cfc3d54777c38edc8782d1abef1087c915ef3800414902cd234	0
types/src/account_config/events/transfer.rs	1627	d9fc333437a3c74dce22510c23b6b85163934bc1b07e12c591ffd38df39ac2e8	0
types/src/account_config/events/transfer_event.rs	1663	24b769e37a7cc7b048ad20436e88819f7a6bdcacd37912742620b367f7d522ce	0
types/src/account_config/events/uri_mutation.rs	2001	a2033d3da1291df0657edfbfd3f6b2c329acfdea5fda8c052285d4b1ac0e7cf3	0
types/src/account_config/events/uri_mutation_event.rs	2037	5c413788631c906e373674bb2c7d275f16f2ec8f2c8afa37a26d466b5fbc7500	0
types/src/account_config/events/withdraw_event.rs	1386	a10fcbf44e4d70cc155aeab757e9d72e25b232c1cd2d8652f4ad21dfa0d09d19	0
types/src/account_config/mod.rs	294	069bf0839db38747fa1c078b13a3237ee38871e0293f20c4354afd4278d60331	0
types/src/account_config/resources/aggregator.rs	2079	2a823ffa3b715d233a4284fe09802e39006ca560032bd945e4d3ffbb9bc7a0ad	0

types/src/account_config/resources/any.rs	1182	59a0d430f860cb217f83f408f79e02e081baa3924565c37577b97b1daf99c939	0
types/src/account_config/resources/chain_id.rs	936	46581f07a365c5b892d71651d5f4b3a5c2712ef832496542721472b3776c444e	0
types/src/account_config/resources/challenge.rs	1143	bff4e473aa8ae9967397ff315c150661f2449eba8ba75f17b9b4da69961515e4	0
types/src/account_config/resources/coin_info.rs	3661	be64345b9157cc123d35ceae80d41c097ebd48a85191d54c2e2d8031d104c008	0
types/src/account_config/resources/coin_store.rs	2804	c23f3df613e8b36b4f3414fd0b1819beaf5119c4af91706a78b6ff0250adf7a0	0
types/src/account_config/resources/collection.rs	1552	7c3829f3ce65b764253948401442ffe5d2a23ac5b31efbec6fa5e774d4824bd7	0
types/src/account_config/resources/collections.rs	1775	ff1957ea66c93012b4c003b754a34976ea383d5a7728496a29eabd973a9546cb	0
types/src/account_config/resources/core_account.rs	3492	976192647c086fcd8bab104396679da6ef1563166713863e8ecdb72b6a11c64b	0
types/src/account_config/resources/fixed_supply.rs	1568	6cf325e3053de8b7b42477df4ebfd46a4dc9090427cea54e4bd3d56ec9efc617	0
types/src/account_config/resources/fungible_asset_metadata.rs	752	31fea1b80b4ecf16e4660b19946f61cd10d1352247ed46a8c248b1b358940fc6	0
types/src/account_config/resources/fungible_store.rs	2944	c1065e321eee5c0914fdef5f7df59c7814e4df86f960df9687b70e36fab9f4bf	0
types/src/account_config/resources/mod.rs	1038	ff09916e3eb34d268d84f0298f642536d7c3166c33801d7f8e14c379a2be54d5	0
types/src/account_config/resources/object.rs	2529	67b5c54c89652c17c90a8f01b3f40dea3c1b3029a02dc04c5b1110cb77b15eae	0
types/src/account_config/resources/pending_claims.rs	1523	7094138db58b287b2c69bf85f902279fdbe6b5d453e55325a5965d95d677d5bb	0
types/src/account_config/resources/token.rs	1626	798de48d5a2808531f4ee818c550a6f8786554bbe4aad6c1f4840f6c10f993f1	0
types/src/account_config/resources/token_event_store_v1.rs	3168	91226223b3778d5e9062946f7321091e58fd30dcb531ef5a74f953de5eefca6f	0
types/src/account_config/resources/token_store.rs	1978	04f0284e9b47b37ac868d72d42c6b0da4925bd5e40bd4c7b0e99353be72c9e76	0
types/src/account_config/resources/type_info.rs	1365	445e2ac3054261f058c66b798ddf94d8b853925215693db88aa05c6b483e5a4d	0
types/src/account_config/resources/unlimited_supply.rs	1449	fb868e5477a052260fa20dadf8a49be5e321911d5c8c9d66b6b15b7b97e0360c	0
types/src/aggregate_signature.rs	4701	7393bfbbb65bb784d967dffee8ef94d02181b7635a82ffb7e8a6e639286f0ba2	0
types/src/block_executor/config.rs	10057	7e76b1a2863363523e44023e3ae1baf3492edc130cf330a43205ed37004ca79d	0
types/src/block_executor/mod.rs	280	ed1ec89a20802b3438c1c274cb86fbe48ec6337f0fd52fc0b66ca52ff924921d	0

types/src/block_executor/output.rs	2405	a33715e27c11fcacd7c29dd16c2791d9bedcae8e538eb9d8ec742fe65f127f7f	0
types/src/block_executor/partitioner.rs	18596	a2fa5d1682231e46bf00c0205f9fb7e7a2a5168f5533083e3cfa24cd85f3e422	0
types/src/block_executor/transaction_slice_metadata.rs	6098	095be58d8c8f719019ee32ed4e043415c750c86c4058e71b35125b9a08880fd1	0
types/src/block_executor/value.rs	4333	1e8e1033ace6235a0415bd0b3d227b2b669290007ce2051636be520f51483b41	0
types/src/block_info.rs	8273	a842d82896dc618bdc9c6e0cff7035a9b4768130e54f8bbd72a78d2b3b7d650c	0
types/src/block_metadata.rs	3333	db388a5a2d123f5155d226dc08ebcf0e6369e16264f252c78d5d6c4283ceed1d	0
types/src/block_metadata_ext.rs	7575	4ac13141cf2f5600fffa9223986b0cc0bac9e681c43c9953c253e2a9581b2aa9	0
types/src/bytes.rs	399	bdce4dd1ab649d5a46d3fa3368b868ee8cdd6b6213dddd2c4f556944aeaec00a	0
types/src/chain_id.rs	7625	22df94b24040c9eefeb02b01fed59887338dd08b37243f2a94ef7da12ae123b3	0
types/src/contract_event.rs	13940	d93e983c084ee74762681eaca670fb908e81869fa2b5801d1d483b8d241ed677	0
types/src/crypto/algebra.rs	12420	757bba4179583778fb17631a7fcf97910694f663bbbd7d7403141265745f4f14	0
types/src/crypto/mod.rs	275	d7a97e463e3edff1c6ae4cb29cb88093bb5ee9104f001c58ba8824b90e9969d9	0
types/src/decryption.rs	3374	455b57c4969a59466028d7665799ec5668112921d3ae99fe670c10be30d8675a	0
types/src/delayed_fields.rs	4514	e78460e1644c67037bef3315f6ade600d7be93e0567f378544d1ec1b7a64fb4f	0
types/src/dkg/chunky_dkg.rs	22794	825c5797d23258c038b228dd800f1df42c3832dc1578d42eb0d45a9ed9175c49	0
types/src/dkg/dummy_dkg/mod.rs	5062	cd6236d967f1937866751af820d09a3430c7c5b648a2536c3797b22032f72bd5	0
types/src/dkg/mod.rs	539	63f2c96d28c8fbee7d76365619712776465f4e024a6db5ddadfc97b12f0956d3	0
types/src/dkg/randomness_dkg.rs	7929	45f99df2a9c913686a82ed5495b83284ed8379cecea4c3c3781f62877cf45684	0
types/src/dkg/real_dkg/mod.rs	16211	1a4dfedefc039e982e0e825e945abe240c7048829bfbbfe6aaade2b3af4c0fd2	0
types/src/dkg/real_dkg/rounding/mod.rs	14699	8093ddc9417f1e1c71a91a92a60d49058e0a57b88662ef394c4c605b8f0e9bad	0
types/src/epoch_change.rs	15554	1dace03de73eddd86587af745c5d10cfa4021e6bf2a6f286e9ab8bf8180957b4	0
types/src/epoch_state.rs	2195	553785c9bc90679e7fa4c54951e60787ae0229891296fabc8a8aa479ab3653c4	0
types/src/error.rs	6934	f3941c855e3d6e88a479b1dfac773b9c610a3616f791ba06cd13452e76c714b3	0
types/src/event.rs	3336	a8aad7b38c093ed895437bc8afc374509d5312dc69b5f14bd02a1a35a3bea9bf	0
types/src/executable.rs	974	d96823fbceaae0567772910dea3c8b7b8d108816eb4b6b18c23334bfd9885ca1	0
types/src/fee_statement.rs	3976	3db2c9f5c8089bd203561395faeacf480708e64f96643c67921a43390ab9e324	0
types/src/function_info.rs	2070	6ba4fec2c2f5b80a17ba6be6e6ae0c0677a62c4bc712a800004bb46a08dcbb5f	0
types/src/governance.rs	359	b7e1339a36e98d037cf65d56964c07deca497b7999228410950045dd98011e1f	0
types/src/indexer/indexer_db_reader.rs	2500	c835f4f244b40c8cceb2bd6cc1498caee03d0663b9c697cf037c184deb52e62b	0
types/src/indexer/mod.rs	203	c0f455b28784da3333bf2a8b36177ba5f4c2a1d2c0ddd0ecc448ed8e5fbcf336	0
types/src/jwks/jwk/mod.rs	4028	604662d9550e09ded059a2e30569f13d82392e447f64cd5d1bd82f7f55486999	0

types/src/jwks/mod.rs	15517	dc31f041e311baf548ee741823e000ccf6c82a6e8a69d1ec3fd3a5aca697d940	0
types/src/jwks/patch/mod.rs	1233	1fe0d1ff6faddebce8e774199123b32a08b1d8c6389677d33f1bd96732f03716	0
types/src/jwks/rsa/insecure_test_jwk_private_key.txt	96	20016265e8d12b02cde1ddbe782ce70e944d46e0d546bcc3bb99c45d10073143	1
types/src/jwks/rsa/mod.rs	7011	f5389d0b7d210a3ce2ff377055ee7f2352c48d9893cf2cf8cf72f586e234bd15	0
types/src/jwks/unsupported/mod.rs	2238	c8a1faa6698059932f1b89d205af3d80ee931971890fe94ee88e17144e746465	0
types/src/keyless/bn254_circom.rs	16339	8493c75c555dad296ad5867c5bc8ae58fd14bc3fb253d2c6b665aea39c359743	0
types/src/keyless/circuit_constants.rs	4443	4e47662a9faa45b82f150000e9557dedeb476c0b1a13fb68fdcf589f9b15ec8b	0
types/src/keyless/circuit_testcases.rs	10671	792305e4068a22b3a300c4734f5f2d27b29f506f04f1f3107bf5bb0872e216e8	0
types/src/keyless/configuration.rs	3566	63b2c1dda971d46b27abefaf0100e9e47558e74d518b5376b185fe15d787e606	0
types/src/keyless/groth16_sig.rs	8306	9de9c6410d73e6d2f7bd990b378957661e189b58e84c4e899b18d6cb3643ed8c	0
types/src/keyless/groth16_vk.rs	5826	9f69a4e8d6e56d2ae1a57ecdb9473b0429f0e426bf79654e71efadcb54ce7c41	0
types/src/keyless/mod.rs	18837	aec84ffbda933e5e148001155cd65cd20890ad13d6bacadc89818ce16bd5f232	0
types/src/keyless/openid_sig.rs	8562	512e5e37c795208f20fe6049ebe68b15b2f73589ef1fe7063910d351ce69343e	0
types/src/keyless/proof_simulation.rs	14668	bf98dac69c4c14a0f2e54b6a3a3d363118a1606442835cb675bf33ab3530b50d	0
types/src/keyless/zkp_sig.rs	783	6395f99ff356f33bd4f10ab26c8270a7b71fd2532f533a43ad93d49c2d10adc3	0
types/src/lazy_bls.rs	14480	1584138e4a7bbeaf8b407a86cd657a87520bd82f05d5c77134374f1983ae0985	0
types/src/lazy_crypto.rs	5590	da03c7bb79b5be9529c23cdd322ed24058c8084e63680ededc47cfcbf993834f	0
types/src/ledger_info.rs	41824	092d6395d8a0ede0b4a356d883abbd002df46083fc07d011cfad0f36e3a8f7ea	0
types/src/lib.rs	1743	43dbe71d6eaadbd8c370ec02cd9b22800eb3a8c02da4de0018dc80e978462721	0
types/src/mempool_status.rs	3205	dcd3837a4bf1028567252be0fd62785499f28d40104301b5cab09fcdf324a8c8	0
types/src/move_any.rs	1494	48f5efbb3faa42b64509bdd33fe525fe8b0713d36a2b1ce0979bce91b6c40fcd	0
types/src/move_fixed_point.rs	587	511cea51c9a17245c51b385ee3fb92732769b9459d5e0b5fb90fbf48c2c9a7f1	0
types/src/move_stdlib/mod.rs	337	0162dd9ea9a9fcc719d1964b038c07ac08aded852f7e385f73312da39e8171b6	0
types/src/move_stdlib/reflect.rs	1335	a6366318338d301c6b7e542668065926c106d3b33c249ef355ec6c5aa0ff6aa1	0
types/src/move_stdlib/result.rs	297	b0069dd5fb5ce823d2d19173e3d96b0c1bf7277e8cc2419e9bb7ddc65f85013b	0
types/src/move_utils/as_move_value.rs	1550	b5e42802a0b157bbb7979eb395779268a3c7bb75ec603922f0bf051e8bd0f03c	0
types/src/move_utils/mod.rs	1409	8683e9930a62fef5362f0261336ae3a0af1d6acff56fd0be6ce22594fbef1ef2	0
types/src/move_utils/move_event_v1.rs	796	bee404230e9a9bce07c6093b12a1b017670251e195940e07ecf1edc7fd6cd87e	0
types/src/move_utils/move_event_v2.rs	585	c6410bf68680dd2c293d00f0d44caff97dd684dfd888603be124ba292b4ebe83	0
types/src/network_address/mod.rs	36409	4975d0e2ec175a34a40327721910a12733f99070948a0e1a9db1742b98a09853	5
types/src/nibble/mod.rs	1676	bcf4607f4d63f6d2c8bcab777e765bdc028deeba411199f9b760525f48932785	0

types/src/nibble/nibble_path/mod.rs	12503	2311ed09285a54b2ecef5b1ee2a053b58f7df3db342283e48855cb739cbe94f8	0
types/src/object_address.rs	733	b5adb0735fe5d0fca5f5c9bd155adfd9bc9620c3db31367ecc9d725a6221f2d0	0
types/src/on_chain_config/approved_execution_hashes.rs	998	dc3e031a1973c42f3d6b1de92c32ea0ef51ed5492245a66d520a88d6758e1c4e	0
types/src/on_chain_config/aptos_features.rs	30005	cb348034ebdbadca299ac0a6173f86585bfa2a803d2fe48a9c78cd6114296bde	0
types/src/on_chain_config/aptos_version.rs	1293	252460a676db6df62688db12f0e5cf65f2dbc9b8cf36fa72a3d8dc5e0e5bce5b	0
types/src/on_chain_config/chain_id.rs	385	b7263da4a34565e6af67f361ce7faa60b661ff0829aeb5412d895bee9cfc2d5f	0
types/src/on_chain_config/chunky_dkg_config.rs	7460	32839bcf2c6cf9fe2304fb083d02dbabfda6848226fd557fa9cff06a1f76296d	0
types/src/on_chain_config/commit_history.rs	1076	2309c5e397aeb5547ec1f98b6372b1285690055ac6c1b987e0a23d4fb2caefa6	0
types/src/on_chain_config/consensus_config.rs	22110	21a10c32c9ce9ce92e0efc096eb803944c8cbc71a4be17173a2518ca1169ce60	0
types/src/on_chain_config/epoch_timeout_config.rs	1375	258a66fea4fd2b8c755582004f626e414d901cef596adedb7574c609f874a7af	0
types/src/on_chain_config/execution_config.rs	21780	89c1b5b455c234b99c8b94b574910fb5349a091b3b035cab054abd9ba9a34b1b	0
types/src/on_chain_config/gas_schedule.rs	3299	66d346bb0d05329c64f10a552180fa8502a27fc11f5bf3509fb595e2d742ef40	0
types/src/on_chain_config/hot_state_config.rs	5812	04a8f61991754c704fa2cc6e421b109a912ace5a83c23c0614a3bf3b55936ba0	0
types/src/on_chain_config/jwk_consensus_config.rs	4353	e4e99234695856703e9bed9d26df4bb77a8788d9cde372783c0621b29ae5441c	0
types/src/on_chain_config/mod.rs	10154	727ceb6f8f2925fc5c5de50570327b3fc6f9bd7c5019bc9293c71574c7a6d912	0
types/src/on_chain_config/randomness_api_v0_config.rs	1661	03eb08ca61b212a02209feafdc03d768611b38d1078b358fe3c29835aafd5749	0
types/src/on_chain_config/randomness_config.rs	7931	3b8d9f3396a7824ee4e71796cbd206424e3743f5f2945e9916806b9cc34543ae	0
types/src/on_chain_config/timed_features.rs	25211	cf14230f8b9407a9f7f8531230ebf8a9d6f37f9f90d52d6b4a6179f5e8392297	0
types/src/on_chain_config/timestamp.rs	564	9b9828e5d45ce4ca8b9eae722baf8030d5b3164a9235e14b40a97e442069beed	0
types/src/on_chain_config/transaction_fee.rs	534	fe9c48693907de4f0153bded4d9d2fa0d4650eff54921958511656b11c8df917	0
types/src/on_chain_config/validator_set.rs	3113	8566fa8178e7b1c8022afb6b0254b0f9c01398a30a93776c99f06d58786811df	0
types/src/proof/accumulator/mock.rs	12581	e94346c424accd0c2e7b3522f037dd21b7ff41b08ae802948fa79be33b893e02	0
types/src/proof/accumulator/mod.rs	15444	a23b47e6f3625080c47d4f9eab21977633df847a59e8928b9d9905f3cdae9b9d	0

types/src/proof/definition.rs	37684	208c571bdaad6894a92892eae79bbe3fe19a4f6cce39fcea0c3ea16df00e1db2	0
types/src/proof/mod.rs	3775	703ba773bb02adde8d2d052827caae891bd4b295f23f92e5504af0c96d5575c3	0
types/src/proof/position/mod.rs	18304	7a3200534d3829481fb9e3448017207a67ce075537b7a101f8814bc8e427ba9b	0
types/src/proof/proptest_proof.rs	5430	b81799ba3c19984b42acb560ccb85147d07f765056873e7c5f4fdfb5508a1976	0
types/src/proptest_types.rs	45523	8f72d7856db3e71d1d5938d3a4be9ea2ec9ddede1ffc3a640557dad324c072a3	0
types/src/quorum_store/mod.rs	1611	48b8fdec0d02837e434eb93216933528247e266bd2008786de1c8f24e4c89cc0	0
types/src/randomness.rs	4002	0b85879162c6fab76de3ee0045520fab66319e5d4f20b6135340c0dab4741f02	0
types/src/secret_sharing.rs	6979	cc7123acff99e51126a5d1ba7f3f7d445cc1b3463603641a6aafc18bef9be1a2	0
types/src/serde_helper/bcs_utils.rs	1003	3fd10762a50f266e9d5a0f8dee245b16f64affa1051477df379a2516766bcf0b	0
types/src/serde_helper/mod.rs	214	fc197ba018cf7f78fc8c32a40f707f4689afbb592349482860ab5521c55c65a7	0
types/src/serde_helper/vec_bytes.rs	816	3f90b12f602af5408b032cc3a99e37909edcbb8ea0608edc693aef34b85f2f70	0
types/src/stake_pool.rs	3360	cc81818fb29765d04a365ef2d45f1f198fcf00e2309cc3effb49b590d363e22e	0
types/src/staking_contract.rs	1574	9c126979748aa93f21bf9912517cb8296d21b48e8fc34b0af869ec8610e82e0e	0
types/src/state_proof.rs	2319	b963f76385b9e6928b234337158a2eb23714885235c04f1f832891e20637ee78	0
types/src/state_store/errors.rs	580	0b8abd23aa009e47cff1ae0480e9fc6c59062655334118d8f6fc82f1e6bd50f5	0
types/src/state_store/hot_state.rs	5855	73de17ceef0d54f28d4bb854d01267ea8dde7766de6fda03dfe424d88d3faf23	0
types/src/state_store/mod.rs	7167	1eaaee36c1cdbb979fc3bb2b7909affe27cbd1308abbf3da41d64225818cde67	0
types/src/state_store/native_collateral.rs	7379	89012a8a387cc3e784dc8337d93bdf3f66c4537b2f39583a7b1be0f98ff942ca	0
types/src/state_store/native_order.rs	25001	8bc7ffe26c5a4cc4a43bed10272d495d22dd0c4372e8101b13e183b1129ef52b	0
types/src/state_store/native_position.rs	5208	ad7eefbe41bbf2c94ea8525b70a15c903593d4402764278e0374734822e0d214	0
types/src/state_store/state_key/inner.rs	9976	37c10fb9b52d27ecc6e17d6ddd4e7169edb6026a5f3587044606dee925e31b86	0
types/src/state_store/state_key/mod.rs	20535	903390ef5c9a3af2ddb9a120b9b9f872bd31e544785f91b4aee01596c894f748	0
types/src/state_store/state_key/prefix.rs	2617	1332a959b0ec7404e1495a76146110c2cadcb45b7571076ff3ae58f7ee812488	0
types/src/state_store/state_key/registry.rs	13622	c45ee16c8f31c18e5a84862714fbc0e4bfcf00bd8d4623a64c66415e3fa51bff	0
types/src/state_store/state_slot.rs	10300	89e8c250501930cb6498aac18b7fb54bbea812e331dc856f01cfd84504de930f	0
types/src/state_store/state_storage_usage.rs	1627	9a1c79a9e14e50f02fa3326873374a09f02f3df2e9e4038a68b83c9fb73588f9	0
types/src/state_store/state_value.rs	12200	9dcce2dd86ededde31650855a6c9dec6e67840b85345a2939f880d96b56dc509	0
types/src/state_store/table.rs	1423	38c815170f3bb244ef44450185300b1a9079369f0dfdf7b6d50fd5f9989aff86	0
types/src/timestamp.rs	778	86886ea56b28a95ace95f0a300a088dc5cd19c4d745161684a6327a14ed0de30	0

types/src/transaction/analyzed_transaction.rs	10548	29d40e893c1de3c399de81fe021c4f049f586896eed62039606a8b71625d8d98	0
types/src/transaction/authenticator.rs	97841	30fa99d247f36e2e359534968093be3d21c9dc3c147ba4c2020144da0652b6c3	0
types/src/transaction/block_epilogue.rs	5362	7cb091e80364786b45fafd187c76b40e00f079682fba76b5ce6a9ff5690cd58f	0
types/src/transaction/block_output.rs	2040	4c805459fbcd14aad7d4795493457e7d113510170f6e438b7987284bfc97fd5e	0
types/src/transaction/change_set.rs	950	44b85b95f9fd0e6d8ed9470b63be690e49b570219468b76c0570a34c01047c7e	0
types/src/transaction/encrypted_payload.rs	12484	f0591e21df64c7f8642119302e13c9e0ebe87ccae3183e11662a53ac11065d34	0
types/src/transaction/mod.rs	132928	88fc7d10033dd069293d85da58869b5768149ae6c8d1138de171781f12572a8e	0
types/src/transaction/module.rs	2146	c2d003708fd6ac4da0bca54aad7799e97bff616caac415273ee00a6388259d79	0
types/src/transaction/multisig.rs	4557	a4eecb859fa4627155af87d52f4894cbeb3de06c667a4938dbf055130402ae95	0
types/src/transaction/script.rs	4371	2a2027fe377911e700b25c4f74e44ac5faaad7576396bfaa8389c403a7eeec47	0
types/src/transaction/session_id.rs	8559	cd1f829c9cdae939ff0b049ee09974cad22a68c6d5b78dd9a9b1602fd3f67fd2	0
types/src/transaction/signature_verified_transaction.rs	5530	e884a43c33d3a2d34e6bd32ef1e6bd3ee1b395c05bd05c2ba55472668f66cab9	0
types/src/transaction/use_case.rs	13583	eab572a6548aba2c0099b87faa170654304acdaa6e61134d4aae5203d8752a58	0
types/src/transaction/user_transaction_context.rs	5042	3dd946a13cca89a2bf30909b1f0e72621dd14a6c5d2b68f110c18981e7bd6daa	0
types/src/transaction/validation.rs	6144	eaa8a946ac92f3bf40fe1d5dcf028300a6219381987cae99a98c8c92a4ae3d06	0
types/src/transaction/webauthn.rs	43853	27b8e1351299b43f17df226ebd75acae01c352c824f045d3006ca5cf9ba768f7	0
types/src/trusted_state.rs	11468	890651066fe0b81518f4d4b9304d4739a168fd0575384bce942ac9b4f469da7f	0
types/src/utility_coin.rs	1739	f1e11675a7d0f3f9e1c47525fb3f1f6f61f65e04582f0b5afb978996357d3d04	0
types/src/validator_config.rs	2303	f7297cbdd6447d0009df70317ab2882a5dd8b01313f853e4907a5f56e5cf4f4f	0
types/src/validator_info.rs	3442	b8ceaf28929b6d76e43092874d29dd4fd6950d994322b5bb757e38512ef36e00	0
types/src/validator_performances.rs	527	45f92eacfbe85e7887bbfe5fab4b1430dda400af581323b98d524352efaf764d	0
types/src/validator_signer.rs	5137	526b657d7e6298a200db93226ef00db6d7eda4c99d7c5bbcd446393a196338ed	0
types/src/validator_txn.rs	2479	fbcbe49c93c332928e5d6ddeaab2ebd576140f9ab27b88432fffe0d405c61729	0
types/src/validator_verifier.rs	45626	e9a43a02fce8923b4a650ae5b7766dfd14edb2bdeee9887868efcb6635937aff	0
types/src/vesting.rs	450	00519bc61fd556235206c390d5c20ad62a2184b72b00077b5cbbcaefb390f72d	0
types/src/vm/code.rs	943	25cfc228df49de3e7427822ab05d779d5b6c02135dc45e7635282a54f12bd394	0
types/src/vm/mod.rs	232	087e680bcae319fe803bf60f867c5aeacad8234451f1e5e30ffbbf82a327d671	0
types/src/vm/module_metadata.rs	28943	f92b6c5476a0da302daf1222d9c5007aac863199d649fffe81703584d29f5abf	0
types/src/vm/modules.rs	1420	898742b2db80c5d82bf5d624867a14c93dcd9c915322463b4b9f2c8222f0e56f	0
types/src/vm_status.rs	335	078c5919053e6d1cc4b390c0f89ca338d0563be8cb35ec2d80fba4bb78580a68	0

types/src/waypoint.rs	7283	371de942d0930a34581935c4ac53fb3801aeed12aa65acbe62c4f1bfbd74c899	0
types/src/write_set.rs	39838	0198b5622e7ba3de97474e0771f65b2403fe45e5eae3897ef0de1ea66056266a	0